

Quiz 1

Course: Operating Systems

Section: BDS-5A

Name: _____

Course Code: CS 2006

Total Marks:10

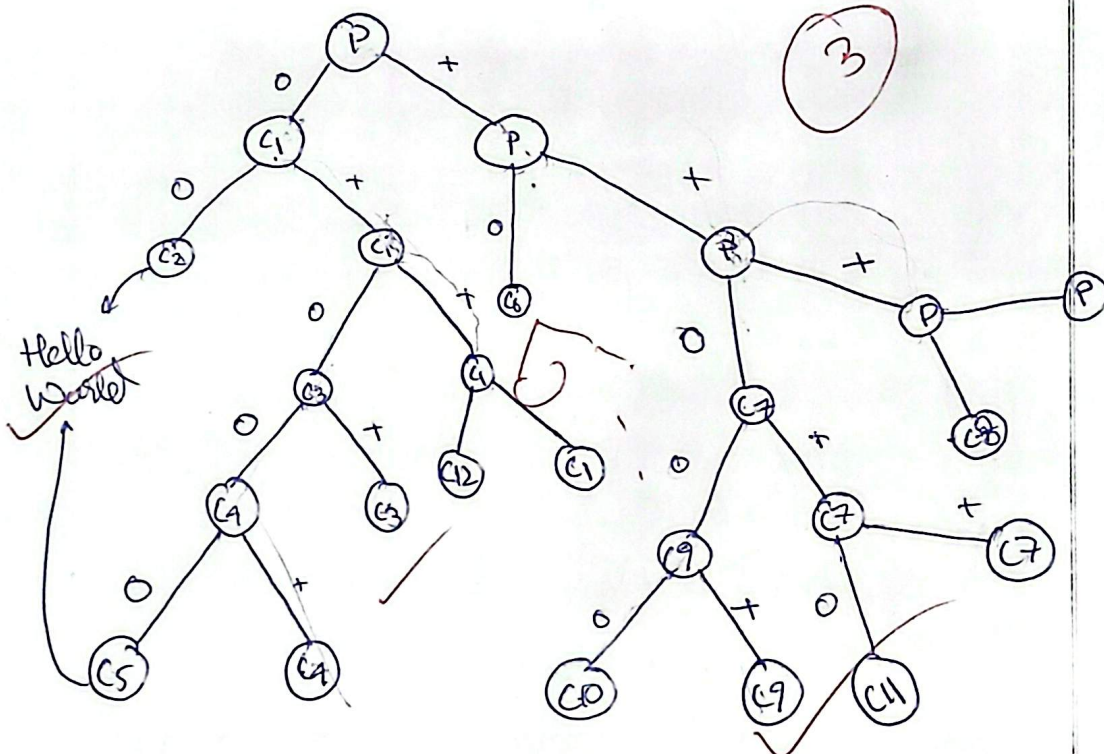
RollNo:

Question 1: [5 Marks] consider the following code:

```
if ((fork() && fork()) || fork())
{if( fork() || fork())
{      fork ();      }
else
{ fork();
}}
printf ("Hello World");
```

How many times "Hello World" will be printed on the screen? Also draw the Process Tree of the given program.

Answer:



13 Times

Question 2:[5 marks] A program needs to transfer 32 KB of data from disk to main memory.

- Each disk block is 512 bytes.
- In interrupt-driven I/O, the CPU is interrupted after every block is transferred.
- In DMA-based I/O, the CPU is interrupted only once at the start and once at the end of the whole transfer.
- Handling one interrupt takes 200 instructions, and the CPU runs at 1 million instructions per second.
- How many interrupts occur in interrupt-driven I/O and how much CPU time is wasted on them?
- How many interrupts occur in DMA-based I/O and how much CPU time is wasted there?

Answer:

→ 512 Bytes per Block

→ 32 KB = $32 \times 1024 = 32768$ Bytes in a Disk

→ $\frac{32768}{512} = 64$ Blocks need to be transferred

~~→ 200 ins per interrupt~~

1 interrupt per Block

i. → So: 64 Interrupts in ~~I/O~~ Interrupt Driven I/O

ii. → 200 ins per interrupt

→ 10×10^6 ins per sec

~~→ 64 Blocks = 64 ints~~

~~→ $200 \times 64 = 12800$ ins~~

~~→ 10×10^6~~

~~→ So After 64 Blocks → 200 ins~~

→ $\frac{ins}{ins} = \frac{1}{s}$

→ $\frac{12800}{10000000} = \frac{128}{1000000} = 0.000128$ seconds